

REPUBLIC OF CAMEROON

Peace – Work – Fatherland

MINISTRY OF SECONDARY EDUCATION

INSPECTORATE GENERAL OF EDUCATION

Inspectorate of Pedagogy in charge of the
Teaching of Computer Science



REPUBLIQUE DU CAMEROUN

Paix – Travail – Patrie

MINISTRE DES ENSEIGNEMENTS SECONDAIRES

INSPECTION GENERALE DES ENSEIGNEMENTS

Inspection de Pédagogie chargée de
l'enseignement de l'informatique

NATIONAL HARMONISED LOWERSIXTH PROGRESSION SHEET FOR INFORMATION & COMMUNICATION TECHNOLOGY

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 8 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of Action	Lesson Title	Objectives	Resources	Nature of Lesson		
							Th	Prac	Dig
First Term	1	Computing Systems and Component s	History and Evolution of Computing	Diagnostic Evaluation Lesson 1: History and Evolution of Computing	- Identify the generations of computers and the main technology used in each period. - Compare characteristics such as size, processing capability and price across generations. - Differentiate between the Von Neumann and Harvard architecture. - Explain the stored program concept.	AI SIMULATIONS AND AI TOOLS			
		Impacting society with digital technologie s	Exploring AI Concepts	Lesson 2: Introduction to Artificial Intelligence (AI)	- Define AI and explain its history and evolution. - Identify and describe the main types of AI. - Explain common applications of AI.				
				Lesson 3: AI Ethics and Responsible Use	- Explain ethical issues associated with AI. - Examine the potential risks and benefits of AI. - Discuss principles for the responsible use of AI.				
	Lesson 4: AI Techniques and Intelligent Systems			- Describe common AI techniques and their applications. - Explain the characteristics of intelligent systems. - Compare AI techniques and intelligent systems based on their applications					
	Lesson 5: Machine Learning			- Define Machine Learning. - Explain the basic principles of Machine Learning. - Distinguish between supervised, unsupervised and reinforcement learning.					
	2	Lesson 6: Developing AI Systems	- Explain the stages involved in developing an AI system. - Identify programming languages and tools used in AI development.						

			Description of computing trends and categorization of computers	Lesson 7: Types of Computers	• Explain why Python is widely used in AI development. • Identify types of computers (supercomputer, mainframe, minicomputer, microcomputer) • Describe each type of computer with respect to size, power, cost and purpose • Choose an appropriate computer type suited to a given situation				
				Lesson 8: Basic Components of a Computer	• Explain the concepts of hardware, input, output, storage and processing device. • State the purpose of common devices in each category (input, output, storage, and processing) • Choose an appropriate device for a given situation				
				Lesson 9: Input devices	• Explain the concept of automatic data capture • Describe the following automatic data capture devices (MICR, OCR, OMR, barcode reader, QR code reader, card readers, RFID readers, Biometric readers). • Explain how AI-based image recognition extends automatic data capture beyond QR/barcode reading.				
	3	Computing Systems and Components	Hardware and Categorisation of Hardware components	Lesson 10: Output devices	• Differentiate between the different types of printers (dot matrix printer, laser printer, inkjet printer) • Describe the following output devices (projector, graph plotter, 3D printers, actuators). • Justify choice of an output device in a given situation.				
				Lesson 11: Secondary Storage media and devices	• Differentiate between secondary storage and primary storage. • Describe the different types of secondary storage media and devices (Magnetic storage, Optical storage, and Solid-state drives). • Justify choice of a storage media and device in a given situation.				
				Lesson 12: Primary Storage devices	• Describe types of primary storage (RAM, ROM, cache, registers). • Compare memory types based on access speed, cost and storage capacity. • Explain why large-scale AI/machine-learning training requires significant secondary storage capacity for its datasets.				
				Lesson 13: Processing device and the machine instruction cycle	• Differentiate between CPU and GPU • Describe the different stages of the machine instruction cycle. • Explain why AI/machine-learning workloads rely heavily on parallel GPU processing rather than the CPU alone.				
				Lesson 14: Processor architectures, parallel & distributed computing	• Explain processor architectures (CISC and RISC). • Describe Flynn's architectures (SIMD, SISD, MISD, MIMD). • Differentiate between parallel and distributed computing.				
				Lesson 15: Conversion between units of storage and units of processing	• Identify units of storage and units of processing • State the relationships between storage units (bits, bytes, terabytes, ...) and processing units (Hertz, Kilohertz, Gigahertz) • Convert from one unit of storage to another.				
	4		Computer Software	Lesson 16: Definitions and classification of software	• Explain the concept of software. • Explain the difference between system software and application software.				

					Establish with examples the difference between open source and proprietary software.				
	5			Lesson 17: Application software	- Differentiate between general purpose, specific purpose and tailor-made software. - Describe common types of application software (word processor, browser, spreadsheet, desktop publishing, , graphic, Presentation, artificial intelligence, etc). - Choose appropriate application software for a given task or situation.				
				Evaluation No 1					
				Remediation No 1	- Review and correct the common errors identified in Evaluation No 1. - Reinforce weak areas in computing history, AI concepts and hardware categorisation through guided practice. - Support learners who under-performed before moving into computer software and operating systems.				
				Lesson 18: System software and examples	- Describe the different types of system software (operating system, device drivers, utility software). - Describe common types of utility software (antivirus, disk cleaner, defragmenter, File management, File compression, etc) - Identify examples of AI-powered utility software, such as intelligent antivirus and predictive disk optimisation.				
	6		Operating Systems	Lesson 19: Notions of the Operating System	- Outline major landmarks in the history and evolution of operating systems. - Describe types of operating systems (single user, batch, online, multi-user, NOS, multitasking, real-time). - State functions of an operating system.				
				Lesson 20: Functions of an operating system 1	- Distinguish between pre-emptive and non-pre-emptive scheduling. - Describe scheduling algorithms (FCFS, SJF, SRT, round robin). - Explain briefly how AI techniques are used in modern operating systems for predictive process scheduling.				
				Lesson 21: Functions of the operating system 2	- Explain how the operating system manages memory, files, Devices (Input & output). - Explain the following concepts as used in operating systems: deallocation, virtual memory, buffering, spooling, metadata.				
				Lesson 22: Installing an operating system and user interfaces	- Install an operating system (Windows or Linux) - Identify OS interfaces (GUI, CLI) and their strengths and weaknesses. - Select appropriate interface for a given context.				
				Lesson 23: Using the GUI of an operating system	- Explain the following concepts: file, folder, file format. - Perform operations (creation, renaming, deleting, copying, cutting, compressing, ...) on files and folders using the GUI of an operating system. - Set up an OS to avoid unauthorised access using a GUI.				
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				Lesson 24: Using the CLI of an operating system	- Launch and identify command line environments (MS-DOS or Linux). - Perform operations on files and folders using the CLI. - Write and execute a command-line script to automate a task.				
			Hardware and Software Maintenance	Lesson 25: Hardware faults identification and correction	- Explain the following concepts computer maintenance, preventive maintenance, corrective maintenance - Describe common hardware faults in a computer system and techniques to curb them.				
				Lesson 26: Software faults identification and correction	- Differentiate between hardware and software maintenance. - Describe common software faults and how to curb them.				
	8		Assistive technologies and computer ergonomics	Lesson 27: Assistive technology	- Explain the concept of assistive technology. - Describe assistive technologies (Braille keyboard, audio devices, automatic speech recognition) - Explain how AI-based speech recognition and predictive text have improved assistive technology for learners with disabilities.				
				Lesson 28: Computer ergonomics	- Explain the concept of computer ergonomics. - Explain computer-related health hazards and their causes. (musculoskeletal disorders, eye strain) - Describe correct posture, habits, and equipment positioning to avoid computer related health hazards				
				Lesson 29: Editing and Formatting text	- Perform text formatting with a word processor - Perform text editing with a word processor - Identify how AI-based writing assistants support grammar, style and content suggestions during text editing.				
				Lesson 30: Editing and formatting images and tables	- Perform editing and formatting of images and graphics - Perform editing and formatting of tables				
				Lesson 31: Using Text boxes and adjusting Page layout	- Perform editing and formatting of text boxes. - Modify features of a page (margins, orientation, size, columns, ...). - Add comments to sections of a document.				
	9		Using a word processor	Integration Activity No 1	- Given a computer system to be made operational for a stated user, including one requiring assistive technology, select and justify appropriate system and application software. - Install/configure the operating system and required software, applying correct ergonomic setup and safe working practices. - Demonstrate at least one accessibility feature suited to the stated user's needs.				
		Practical Problem solving in the digital word							
			Using electronic spreadsheets	Lesson 32: Introduction to spreadsheets	- Describe the following spreadsheet components: row, column, row number, column number, cell, cell address, range, formula, sheet. - Select a row, column, cell, range, and non-contiguous cells on an electronic spreadsheet. - Perform basic editing and formatting using a spreadsheet.				

				Lesson 33: Performing calculations using spreadsheets	- Perform basic arithmetic operations using an electronic spreadsheet (without using functions). - Perform calculations using the SUM, AVERAGE, COUNT, PRODUCT, IF function, COUNTIF. - Explain how AI-powered spreadsheet functions can automatically detect data trends and generate forecasts.				
	10			Lesson 34: Types of cells referencing and calculations	- Differentiate between the different types of cells referencing. - Solve data problems using appropriate spreadsheet functions (SUM, AVERAGE, COUNT, PRODUCT, IF function, COUNTIF, RANK) and cell reference.				
				Evaluation No 2					
				Remediation No 2	- Review and correct the common errors identified in Evaluation No 2. - Reinforce weak areas in software, operating systems and productivity tools (word processor, spreadsheet) through guided practice. - Support learners who under-performed before continuing with the next block.				
			Digital Citizenship and Ethics	Lesson 35: Positive and Negative Uses of Computer Systems	- State positive and negative uses of computer systems. - Explain the social and economic effects of computer systems. - Explain how ICT can be responsibly used to influence communication and culture.				
	11			Lesson 36: Computer Ethics, Legislation and Cameroon Law	- Explain the code of ethics and moral obligation of computer users. - Explain unethical acts with focus on those punishable by Cameroon law. - Discuss ethical concerns raised by AI-generated content, deepfakes and algorithmic bias.				
				Lesson 37: Data Protection, Copyright and the Digital Divide	- Demonstrate knowledge of current data protection and copyright acts. - Determine the effects of global communication on citizenship, culture and digital divide. - Suggest practical ways of reducing the digital divide in the community.				
			Cybersecurity	Lesson 38: Protecting Computer Systems from Illegal Access	- Explain the security, reliability and resilience of computer systems. - Describe measures used to protect systems from illegal access (passwords, security codes, encryption, biometrics, physical security). - Explain the necessity of data handling and backup.				
				Lesson 39: System Recovery and Safe Working Practices	- State the importance of recovery in the event of system failure. - Explain the importance of privacy and safe working practices. - Illustrate with examples how safe working practices protect a computer system.				
				Lesson 40: Computer Crimes and Combat Measures	Describe data handling and computer-related crime (copyright infringement, plagiarism, computer-assisted crime).				
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					- Classify computer crimes into computer-related and computer-assisted crimes. - Associate types of computer crimes to specific measures used to combat them.				
				Lesson 41: Malware — Types and Characteristics	- Describe different types of malware (virus, worm, Trojan horse, rootkit, backdoor, spyware). - Compare the different types of malware based on how they spread and their effects. - Explain how AI is used both to detect malware through behavioural analysis and to create more sophisticated, AI-generated attacks.				
				Lesson 42: Protecting a Computer System from Malware	- Explain good practices (opening trusted attachments, scanning email, downloading from trusted websites) to protect against malware. - Scan a computer system using an antivirus. - Set up protection against unauthorised access using a firewall.				
				Integration Activity No 2	- Produce a properly formatted document and an accompanying spreadsheet performing at least one calculation, both relevant to a stated real-life task. - Identify the ethical, legal and data-protection obligations that apply to handling the data used in the task. - Apply at least two practical measures to protect the data/system used from unauthorised access or malware.				
Second Term	13		Description of Systems	Lesson 43: Types of Systems	- Describe a system and distinguish between natural and artificial systems. - Differentiate between manual and automatic systems (e.g. email, e-learning). - Categorise systems in the learner's environment (manual, automatic).				
				Lesson 44: Modelling a System with a Data Flow Diagram	- Propose limitations on an identified manual system. - Produce a data flow diagram for a given system. - Identify an AI-driven system as an example of an automatic system that can be modelled with a data flow diagram.				
			Information Systems and Data processing	Lesson 45: Introduction to Information Systems	- Define an information system - state the components of an information system and their role (people, procedure, technology, data). - Describe the hierarchical structure of an organisation. - Identify factors affecting the success or failure of an information system. - Establish the necessity for information systems in a library, hospital, company or school.				
				Lesson 46: Types of Information Systems	- Describe types of information systems (MIS, DSS, EIS, TPS, GIS, HIS, LIS). - Compare TPS, MIS, DSS and EIS based on purpose, users and inputs. - Select a suitable information system for a given context.				

					Explain how AI techniques enhance Decision Support Systems (DSS) and Executive Information Systems (EIS).				
	14			Lesson 47: General and Commercial Data Processing	- State examples of general and commercial data processing systems (stock control, order processing, ...). - Explain how general and commercial data processing works. - Differentiate between real time processing and batch processing. - Explain how Machine Learning extends traditional data processing by detecting patterns and making predictions from data.				
			Artificial Intelligence and emerging technologies	Lesson 48: Computer Systems in Industry and Science	- Describe the application of computer systems in sciences and industries (weather forecasting, CAD, CAM, image processing, simulation). - Explain how industrial, technical and scientific systems work. - Evaluate the level of automation and technological tools used in a manufacturing or scientific organisation.				
				Lesson 49: Robots and Their Application in Daily Life	- Describe a robot. - Explain how robots are used in manufacturing, health and the home. - Illustrate with examples situations that may need the use of robots.				
				Lesson 50: Monitoring systems, Control systems, and Monitoring & Control systems	- Differentiate between monitoring system and control system. - Explain how a monitoring system works and give examples of such a system in real-life. - Explain how a control system works and give examples of such a system in real-life. - Explain how a monitoring & control system work and give examples of such a system in real-life. - Describe how monitoring, control, and monitoring & control systems are used in different areas of life.				
	15			Lesson 51: Simulation and Modelling	- Differentiate between simulation and modelling - Explain the concept of simulation using games, videos and head-mounted displays as examples. - Establish advantages and limitations of simulation. - Describe how simulations are used in different domains of life. - Design a simulation of a situation in the learner's environment using technology.				
				Lesson 52: Embedded systems and IoT	- Differentiate between an embedded system and an IoT system. - Explain how an embedded system works and give examples of embedded systems. - Explain how an IoT system works and give examples of IoT systems. - Describe how embedded systems are used in different areas of life. - Describe how IoT systems are used in different areas of life.				
				Lesson 53: Virtual Reality and Augmented Reality	Explain the concepts of Virtual Reality (VR) and Augmented Reality (AR).				

					• Explain how Virtual Reality (VR) works and describe its role in different areas of life. • Explain how Augmented Reality (AR) works and describe its role in different areas of life. • Differentiate between AR and VR systems.				
				Evaluation No 3					
	16			Remediation No 3	• Review and correct the common errors identified in Evaluation No 3. • Reinforce weak areas in digital citizenship, cybersecurity and information systems through guided practice. • Support learners who under-performed before continuing into Second Term.				
				Lesson 54: Multimedia Systems and Authoring	• Explain the concept of multimedia, multimedia authoring tool. • Differentiate between multimedia and non-multimedia content. • Explain how multimedia systems work and give examples of multimedia authoring tools. • State the features of multimedia authoring tools and use one to produce a short multimedia content.				
				Lesson 55: Multimodal Systems	• Differentiate between a unimodal and a multimodal system. • Explain how multimodal systems work. • State real-world examples of multimodal systems. • Describe how multimodal systems are used in different areas of life.				
				Lesson 56: AI technologies	• Differentiate between machine learning, neural networks, and deep learning. • Describe application areas of machine learning, neural network and deep learning. • Analyse real-world problems across diverse domains (healthcare, agriculture, finance, ...) and determine whether they require traditional machine learning, neural networks, or deep Learning solutions. • Train a simple visual machine learning model using an interactive no-code environment.				
	17			Lesson 57: Generative AI	• Explain the concepts of generative AI and large language models. • Explain how generative AI works and give examples of generative AIs. • Explain the concepts of hallucinations, bias, and outline common sources of bias. • Evaluate the impact of Generative AI applications on working and living standards. • Discuss ethical ways of using generative AI.				
				Lesson 58: Using generative AI for day-to-day activities	• Formulate structured prompts to execute real-world workplace or day to day tasks using Generative AI. • Refine prompts and write effective follow-up prompts.				

				Integration Activity No 3	- Given a described organisation or process, model it as a system (data flow diagram) and identify the type of information system best suited to it. - Propose how automation, robotics or AI could improve the process, referencing suitable examples from the categories studied. - Justify the proposed system and technology choices against the organisation's stated needs.				
		Building ICT systems		Lesson 59: Introduction to the System Development Life Cycle (SDLC)	- Define the System Development Life Cycle (SDLC). - Describe each stage in the SDLC. - Identify activities involved at each stage of the SDLC.				
	18		SDLC and SDLC models	Lesson 60: SDLC Models	- Describe the waterfall, prototyping, Agile, Boehm's spiral model. - Compare SDLC models (waterfall, prototyping, Agile, Boehm's spiral).				
				Lesson 61: The prototyping model	- Explain the steps involved in prototyping. - Explain reasons for using prototyping in system development. - Compare the types of prototyping based on cost, time and reusability. - Choose the appropriate type of prototyping for a given context. - Explain how AI-assisted prototyping tools can accelerate the creation of a system prototype.				
				Lesson 62: Changeover Strategies	- Identify the stage of the SDLC where changeover is applied. - Explain the different changeover strategies (direct, parallel, pilot and phased). - Explain advantages and disadvantages of each change over strategy. - Propose a suitable changeover strategy for a given context.				
			Modelling Data in an Information System	Lesson 63: Data Modelling Concepts	- Distinguish between flat file and relational databases. - Explain the different types of database model (ER model, relational model, object-oriented model). - Describe the different levels of data modelling (conceptual, logical and physical). - Outline steps involved in relational data modelling (designing tables, determining primary keys, modelling relationships, normalise models, model queries). - Justify the choice of a database model for a given real-life situation.				
	19			Lesson 64: Entity Relationship Model Concepts and representation systems	- Explain the notions of entity type (table), entity (record) attribute, primary key, relationship, cardinality, foreign key. - Describe the different types of relationship (one to one, one to many and many to many). - Outline common ER modelling notation systems (Chen, Crow's foot). - Identify the symbols in a given ER modelling notation system (Chen, Crow's foot).				

				Lesson 65: Designing an ER diagram and normalisation	- Identify attributes, entity types, primary keys, relationships when given a scenario related to database modelling. - Produce simple ER diagrams (maximum 3 entities) for a given real-life scenario using Chen or Crow's foot notations. - Describe the main types of normal forms (1NF, 2NF, and 3NF).				
				Lesson 66: Normalisation and converting ER model to relational model.	- Transform entities into 1NF, 2NF and 3NF. - Verify the respect of normal forms by an ER model. - Outline rules for moving from an ER model to a relational model. - Transform an ER model into a relational model				
				Lesson 67: Implementing a relational model	- Explain the concepts of DBMS and RDBMS. - State examples of Relational Database Management Systems (Access, Open base, MySQL). - Represent a relational model using a relational database management system (Access, Open base, MySQL).				
	20			Lesson 68: Using an RDBMS for sorting, filtering and queries	- Explain the concepts of sorting, filtering and queries as used in databases. - Perform sorting and filtering using an RDBMS - Write an execute queries using an RDBMS.				
				Integration Activity No 4	- Given a stated information system project (the one modelled in Integration Activity No 4, or a new case), produce a simple prototype or simulation, including a multimedia/VR-AR element where relevant. - Model the system's data using an Entity Relationship diagram and explain the relationships identified. - Explain how the prototype and data model would guide full system development.				
			Software Testing	Lesson 69: Types of software testing	- Explain the concept of software testing. - Explain reasons for testing an information system. - Differentiate between black-box and white box testing - Describe types of testing by explaining how they work, their purpose, and who conducts the test (system testing, volume testing, module testing, integration testing, acceptance testing). - Select a suitable testing method for a given context.				
				Lesson 70: Module testing	- Explain the concepts of test case, test data, bug, actual results, expected results, and test report as used in software testing. - Outline the procedure for module testing. - Carry out the procedure for module testing. - Identify how AI-powered testing tools can automatically generate test cases and flag anomalies.				
				Evaluation No 4					
	21			Remediation No 4	- Review and correct the common errors identified in Evaluation No 4. - Reinforce weak areas in emerging technologies, systems design and database modelling through guided practice. - Support learners who under-performed before continuing with the next block.				

				Lesson 71: Introduction to data representation and number systems	• Explain the concept of data representation and number system. • Explain how data is represented in the computer. • Differentiate between bit, nibble, byte, and word. • Identify number systems (binary, octal, decimal, hexadecimal). • Convert from base 10 to base 2, 8 and 16. • Explain how positive numbers are represented in the computer.				
			Representing numbers	Lesson 72: Conversion between number systems and representing negative numbers	• Convert from base 2, 8, and 16 to base 10. • Outline ways of representing negative numbers. • Describe sign magnitude, One's complement, and two's complement representation. • Compute one's complement and two's complement of a binary number.				
	22			Lesson 73: Subtraction and representing negative numbers	• Perform addition in base 2, 8, and 16. • Carry out subtraction in base 2, 8 and 16.				
			Building Logic Circuits	Lesson 74: Logic gates	• Identify different logic gates (OR, AND, NOT, NAND, NOR, XOR, XNOR). • Sketch logic gate symbols. • Draw truth tables for given logic gates and expressions (up to 3 input variables).				
				Lesson 75: Logic circuits	• Explain the concepts of logic circuit and Boolean expressions. • Derive a logic circuit from a Boolean expression. • Derive a Boolean expression from a logic circuit. • Produce a truth table from a Boolean expression or logic circuit.				
				Lesson 76: Boolean expressions from truth table and De Morgan's laws	• State the ways of getting a Boolean expression from a truth table. • Derive Boolean expressions from truth tables. • State the De Morgan's laws. • Establish the correctness of the De Morgan's laws.				
				Lesson 77: Simplification of Boolean expressions	• Explain the purpose and advantages of simplifying Boolean expressions. • State laws used in simplifying Boolean expressions. • Simplify a Boolean expression using appropriate laws. • Identify when to use De Morgan's laws in Boolean simplification.				
	23		Developing Software — Data Types and Structures	Lesson 78: Data Types and Data structures	• Identify standard data types (integer, real, Boolean, character, string). • Choose a suitable data type for a simple problem. • Explain what a data structure is and why it is useful. • Describe arrays, records, stack, and queue. • Choose appropriate data structure in a given situation.				
				Lesson 79: Operations on data structures	• Explain how elements are accessed in arrays, records, stack, and queue. • Carry out common operations on arrays, records, stack, and queue. • Combine arrays and records to construct new data types.				

				Integration Activity No 5	• Turn the Entity Relationship model from Integration Activity No 5 (or a new stated case) into a normalised relational database, and test it for correctness. • Solve a stated problem using binary arithmetic/number-base conversion, and represent its logic using a logic circuit or Boolean expression. • Explain how correct logic and thorough testing underpin a reliable database/system.				
	24		Designing Software	Lesson 80: Approaches to Software Design	• Describe the following software design techniques top-down design, stepwise refinement, incremental construction, divide-and-conquer, bottom-up design and modular design. • Select a suitable design method for a given problem.				
				Lesson 81: Applying software design techniques	• Break down a problem into smaller units using a software design technique. • Adapt an existing solution to a new problem.				
				Lesson 82: Representing Software Design	• Explain the uses of unit and structure diagrams (class, sequence, composite). • Describe the Unified Modelling Language (UML) model. • Sketch a structure diagram for a given software design scenario.				
				Lesson 83: Introduction to Algorithms	• Explain what an algorithm is. • Explain the characteristics of an algorithm. • Outline the different ways of representing an algorithm. • State advantages and disadvantages of pseudocode and flowcharts.				
Third term	25		Writing and Testing Algorithms	Lesson 84: Algorithmic instructions	• Describe declarative, input, output, and assignment instructions. • Describe the different types of control structures.				
				Lesson 85: Representing Algorithms — Flowcharts	• Represent algorithms using flowcharts. • Use standard flowchart symbols correctly. • Draw a flowchart for a simple real-life problem.				
				Lesson 86: Representing Algorithms — Pseudocode	• Represent algorithms using pseudocode. • Write pseudocode for a simple real-life problems. • Distinguish a rule-based algorithm from a machine-learning algorithm that learns its own rules from data.				
				Lesson 87: Control Structures — Sequence and Selection	• Write algorithms using sequence constructs. • Write algorithms using selection constructs				
	26			Lesson 88: Control Structures — Loops	• Describe the different types of loops. • Write algorithms that make use of the loop control structure. • Differentiate between loops and recursion.				
				Lesson 89: Sorting and Searching	• Explain searching techniques (binary and sequential searches). • Explain sorting techniques (bubble, merge and insertion sorts).				
				Lesson 90: Establishing Correctness of Algorithms	• Explain the concepts of dry running and trace tables • Establish the correctness of an algorithm using dry running. • Draw a trace table for an algorithm.				
				Lesson 91: Evaluating Algorithm Performance	• Explain the concept of complexity of an algorithm. • Examine the performance of an algorithm using time efficiency.				

					- Compare two algorithms solving the same problem based on time efficiency. - Explain how accuracy, precision and recall are used to evaluate the performance of an AI/machine-learning algorithm.				
	27			Evaluation No 5					
				Remediation No 5	- Review and correct the common errors identified in Evaluation No 5. - Reinforce weak areas in binary arithmetic, logic circuits and algorithm writing through guided practice. - Support learners who under-performed before moving into the final programming block of Third Term.				
			Programming Paradigms and Software Reuse	Lesson 92: Programming Paradigms	- Distinguish between types of programming paradigms (imperative/procedural, declarative, functional, logic and object-oriented). - Explain the relative advantages and disadvantages of each paradigm - Explain the properties of object-oriented programming (encapsulation, polymorphism).				
				Lesson 93: Software Reuse	- Explain software reuse and the importance of library units, repositories and software packages. - Explain the types of software reuse (internal and external). - Explain the criteria for selecting software for reuse (reduce development cost, time, labour, maintenance effort, etc). - Identify popular AI/machine-learning libraries and frameworks as an example of software reuse.				
	28			Lesson 94: Internally versus Externally Developed Software	- Distinguish between internally and externally developed software. - Determine the importance of software reuse when outsourcing. - Explain the advantages and disadvantages of internally developed software and externally developed software. - Choose between developing software internally or externally for a given context.				
				Integration Activity No 6	- Given a stated problem, design a solution using appropriate data structures and represent it as an algorithm (flowchart or pseudocode). - Trace and test the algorithm against sample data, evaluating its correctness and efficiency. - Refine the algorithm/design based on the evaluation.				
			Implementing algorithms	Lesson 95: Language Translators and IDE	- Differentiate between language translators (compiler, interpreter, assembler). - Explain advantages and disadvantages of a given language translator. - Explain lexical analysis, code generation, parsing and preprocessing. - Explain the concept of IDE and give examples. - Outline core features of an IDE.				

				Lesson 96: introduction to coding	• Explain the concept of program, coding, and programming language. • Install an IDE. • Explain strategies that can help learners easily adapt and use a programming language. • Explain the meaning of syntax and semantics as used in programming.				
	29			Lesson 97: Writing code 1 (C or Python)	• Identify the structure of a python or C program. • Write input, output, and assignment instructions in C and python. • Write and debug simple programs in C or Python.				
				Lesson 98: Writing code 2	• Write and debug programs in C or Python that make use of selection control structures.				
				Lesson 99: writing code 3	• Write and debug programs in C or Python that make use of loop control structures.				
				Lesson 100: writing code 4	• Write and debug programs in C or Python that make use of arrays.				
				Lesson 101: writing code 5	• Write and debug programs in C or Python that make use of arrays and records.				
	30		Writing functions and testing a developed program	Lesson 102: Functions and procedures 1	• Explain the importance of functions and procedures when writing programs (clarity, efficiency, reuse, maintainability) • Explain the difference between functions and procedures • Identify core components of a function (name, parameters, output) • Write functions in C or Python that solve specific problems. • Perform function or procedure calls in C or Python. • Explain stack mechanism for function call and parameter passing.				
				Lesson 103: Functions and procedures 2	• Write functions in C or Python that solve specific problems and debugged them. • Perform function or procedure calls in C or Python. • Dry run a function call showing stack mechanisms.				
				Lesson 104: Testing a program 1	• Explain the steps involved in software testing (code review, static code analysis, unit testing, system testing). • Explain debugging and compare debugging techniques (brute force, induction, backtracking). • Test a program using appropriate methods and tools.				
				Lesson 105: Testing a program 2	• Test and debug a program using appropriate methods and tools. • Test a developed program using boundary, correct and erroneous data. • Produce a test report.				
	31			Evaluation No 6					
				Remediation No 6	• Review and correct the common errors identified in Evaluation No 6. • Reinforce weak areas covered across the final programming and testing block through guided revision. • Support learners in final preparation for end-of-year requirements.				

		REVISION				
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	END OF PROGRAM					